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UB80610

Marks Distribution

Attendance → 5%

Lab → 25%

Quiz → 15% (Best 3 out of 4)

Mid → 20% (Exam will be on 30)

Final → 35% (Exam will be on 50)

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Topics

CSE110

List, Loops

CSE111

OP basic → class, instance variable + obj.

CSE220

- Array
- Circular Array
- Linked list **
- Stack
- Queue

→ Mid

- Recursion
- Hashing & key
- Indexing
- Tree
- Heap

→ final

→ final.

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31-05-23

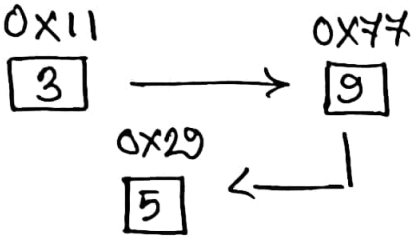
ARRAY

List

1) Dinamic memory Allocation → not fixed, we can append as we wish.

Ex: [3] [9] [5]

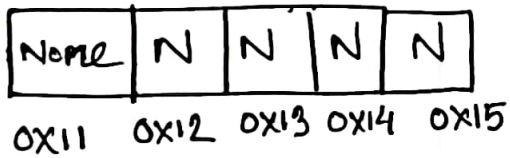
ii)



Random memory Allocation.

ARRAY

- Static memory allocation → fixed size
-



sequential memory allocation

For list,

l = [] ✗

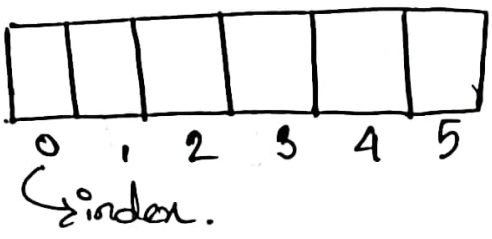
l.append ()

in this course, we want to use append.

(or) `arr = [None] * 5`

Insert elements in an empty array:

```
arr = [None]*6
for i in range (6):
    n = int (input ())
    arr [i] = n
```



Forward Iteration:

arr = [5, 7, 2, 9, 67, 25]
0 1 2 3 4 5

```
for i in range (0, len (arr), 1):
    print (arr [i])

for i in arr:
    print (i)
```
→ the elements

Backward Iteration:

```
for i in range (len(arr)-1, -1, -1):
    print (arr [i])
```

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Given an array, create a new array that is reversed:

rev = [25, 67, 9, 2, 7, 5] } output
      0   1 2 3 4 5

arr = [5, 7, 2, 9, 67, 25] } input
      0 1 2 3 4   5

arr, i		rev, j
0	→	5
1	→	4
2	→	3
3	→	2
4	→	1
5	→	0
i	→	len(arr) - 1 - i

(*) i + j = 5 = len(arr) - 1

(*) j = len(arr) - 1 - i

(**) rev = [none] * len(arr)

```
for i in range (len(arr)):
    j = len(arr) - 1 - i
    rev[j] = arr[i]
```

Two iteratives approach:

```
j = len(arr) - 1
for i in range(len(arr)):
    rev[j] = arr(i)      [out of place reverse]
    j -= 1
```

Left shift:

rev = [7, 2, 9, 67, 25, none] (output)

arr = [5, 7, 2, 9, 67, 25] (input)

arr[1] ⟹ arr[0]
arr[2] → arr[1]
arr[3] → arr[2]
arr[i] → arr[i-1]

Corner case:
0th index
last index

```
for i in range(1, len(arr)):
    arr[i-1] = arr[i]
    ~~arr(len(arr) - 1) = None.~~
arr(len(arr) - 1) = None.
```

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Left Rotate:

arr = [5, 7, 2, 9, 67, 25]

```
temp = arr[0]
for i in range(1, len(arr)):
    arr[i-1] = arr[i]
arr[len(arr)-1] = temp
```

Right Shift:

arr = [5, 7, 2, 9, 67, 25]

0 1 2 3 4 5

rev = [none, 5, 7, 2, 9, 67]

0 1 2 3 4 5

```
arr[0] = none
arr[1] = arr[0]
arr[i+1] = arr[i]
for i in range(0, len(arr)-1):
    arr[i+1] = arr[i]
```

This is wrong!

arr[4] → arr[5]

arr [len(arr) - 2] → arr [len(arr) - 1]

```
for i in range (len(arr)-2, -1, -1):
    arr [i+1] = arr [i]
```

Right Rotate:

```
temp = arr [len(arr) - 1]
for i in range (len(arr)-2, -1, -1):
    arr [i+1] = arr [i]
arr [0] = temp.
```

Rotating n times:

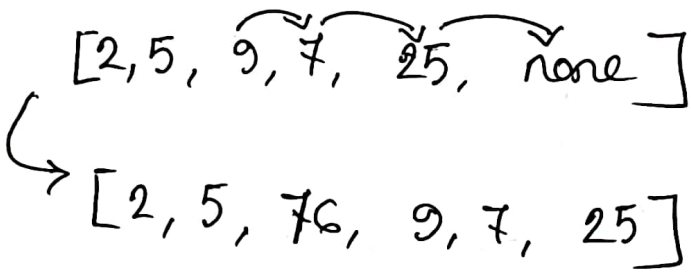
```
n = ....
for j in range (n):
    temp = arr [len(arr) - 1]
    for i in range .
```

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- capacity : Highest no. of elements an array can hold.
- size : Num of non-None elements.

Insert element in index | Index of an array:




```
for i in range (len (arr)-2, idx-1, -1):
    arr [i+1] = arr [1]
arr [idx] = element
```

(*)

```
def insert (arr, idx, element):
    for i in range (len (arr) - 2, idx-1, -1):
```

Deletion:

arr = [5, 7, 2, 9, 67, 25]

in order to dlte 9, we have to left shift.

rev = [7, 2, 9, 67, 25, none] → output

Reverse:

	0	1	2	3	4	5
arr =	10,	20,	25,	16,	7,	9
rev =	9,	7,	16,	25,	20,	10
	0	1	2	3	4	5

arr[0] = arr[5]

arr[1] = arr[4]

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